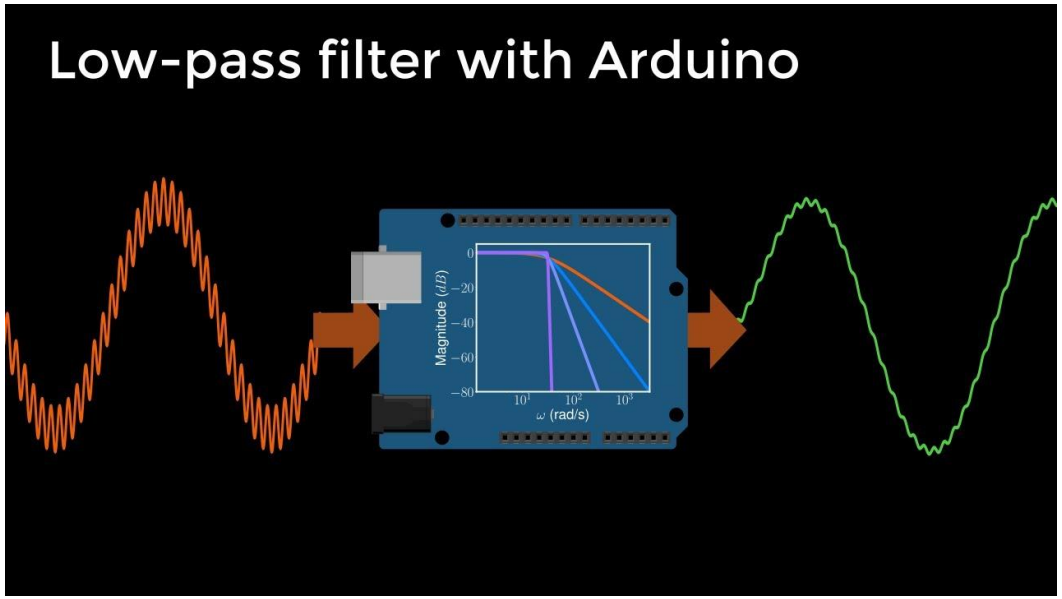
FALL 2026 | MEE4033

Mechatronics

Mechatronics (MEE4033) · Department of Mechanical Engineering



Lecture 9

Passive Filter Experiments

Principles and Applications of Electrical Engineering, 7th ed., Chapter 5
Electric Circuits, 11th ed., Chapters 12, 13, 14

Sogang University · Department of Mechanical Engineering · Prof. Seokhwan Jeong

Why Filters?

- We need filters when the source frequency varies
- Frequency response analysis of a circuit: the response depends on the types of elements (R, L, C) in the circuit
- A circuit can pass to the output only those inputs residing in a desired range of frequencies: a frequency-selective circuit, or filter

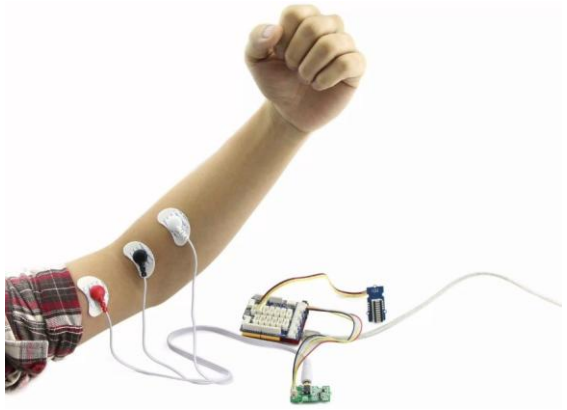


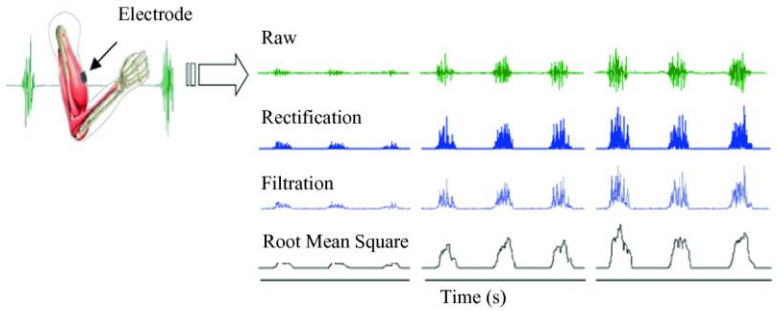
Practical Filters

- No practical filter can perfectly or completely remove selected frequencies; filters attenuate them
- Representative filters: low-pass, high-pass, band-pass, band-reject
- Applications: telephones, radios, televisions, satellites, and biosignals such as EMG









EMG processing: raw → rectification → filtering → RMS. Source: "Experimental Analysis of Electromyography (EMG) Signal for Evaluation of Isometric Muscle Force," Springer (doi:10.1007/978-3-030-27146-6_20)

Resistor and Inductor in the s-Domain

A resistor in the s-domain

t-domain $v = Ri$

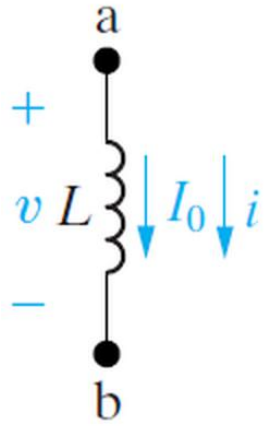
s-domain $V = RI, \qquad \text{where } V = \mathcal{L}\{v\} \text{ and } I = \mathcal{L}\{i\}$



An inductor in the s-domain

t-domain $v = L\dfrac{di}{dt}$

s-domain
$$V = L\left[sI - i(0^-)\right] = sLI - LI_0$$
$$I = \frac{V + LI_0}{sL} = \frac{V}{sL} + \frac{I_0}{s}$$



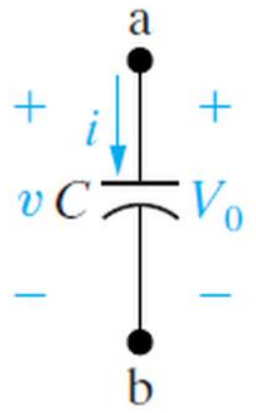
Capacitor in the s-Domain and RC Natural Response

A capacitor in the s-domain

t-domain $$i = C\frac{dv}{dt}$$

s-domain $$I = C\left[sV - v(0^-)\right] = sCV - CV_0$$

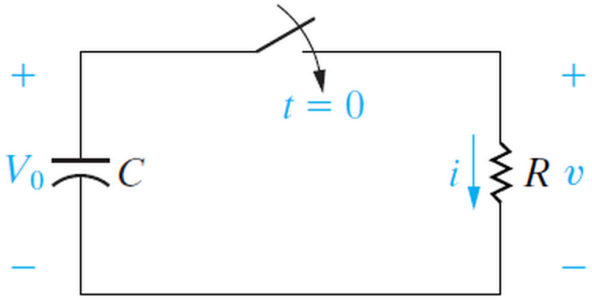
$$V = \left(\frac{1}{sC}\right)I + \frac{V_0}{s}$$



Natural response of an RC circuit

t-domain $$C\frac{dv}{dt} + \frac{v}{R} = 0$$

s-domain $$sCV - CV_0 + \frac{V}{R} = 0 \;\Rightarrow\; V = \frac{V_0}{s + 1/RC} \;\Rightarrow\; v(t) = V_0\, e^{-t/RC}$$

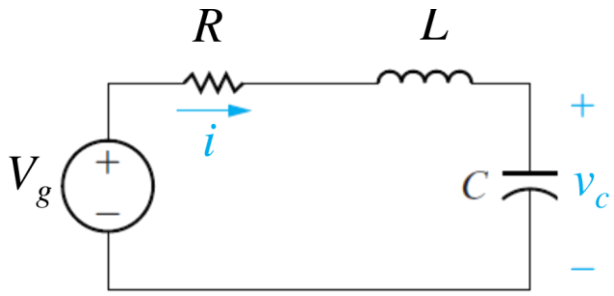


Transfer Function of an RLC Circuit

- **Transfer function**: ratio of the Laplace transforms of the output and the input (all initial conditions zero)

$$H(s) = \frac{Y(s)}{X(s)}$$

Input: v_g / Output: i



$$v_g = Ri + L\frac{di}{dt} + \frac{1}{C}\int_0^t i\,d\tau + v_C(0)$$

$$V_g = RI + L\left[sI - i(0^-)\right] + \frac{I}{sC} + \frac{v_C(0^-)}{s} \xrightarrow{\text{zero ICs}} V_g = \left(R + sL + \frac{1}{sC}\right)I$$

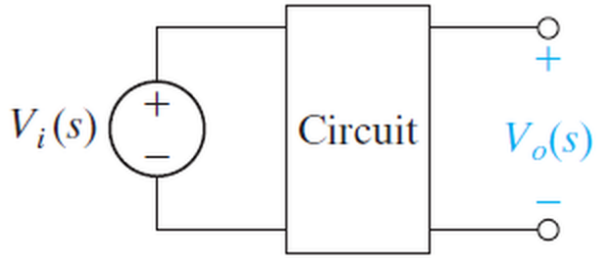
- The transfer function depends on what is defined as the output and the input

$$H(s) = \frac{I}{V_g} = \frac{sC}{s^2LC + sRC + 1}$$

$$H(s) = \frac{V_C}{V_g} = \frac{1}{s^2LC + sRC + 1}$$

Frequency Response of a Circuit

- **Frequency response analysis**:
  - sinusoidal source of varying frequency, **H(jω) = H(s)** evaluated at **s = jω**
- The transfer function gives the magnitude and the phase of the output signal



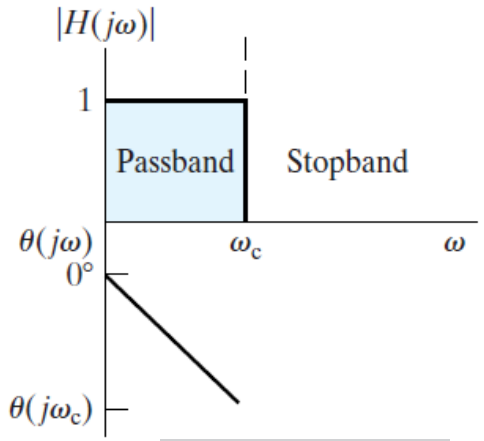
$$H(j\omega) = H(s)|_{s=j\omega}, \qquad H(s) = \frac{V_o(s)}{V_i(s)}$$

Phasor form

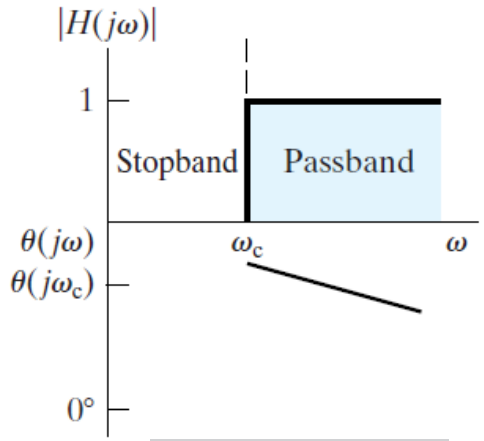
$$\underbrace{M_o\angle\phi_o}_{\text{output}} = \underbrace{(M_i\angle\phi_i)}_{\text{input}} \qquad \underbrace{(M_H\angle\phi_H)}_{\text{frequency response function}}$$

$$M_H = |H(j\omega)|, \qquad \phi_H = \angle H(j\omega)$$

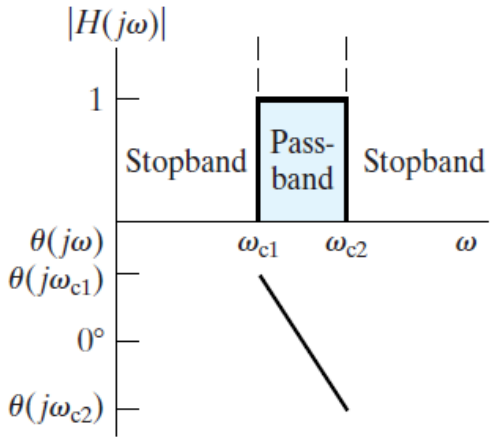
Ideal Filter Responses (Bode Plots)



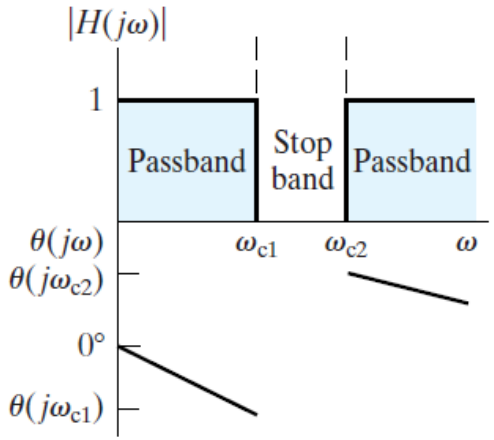
Low-pass filter



High-pass filter



Band-pass filter



Band-reject filter

(These are ***ideal*** frequency response plots for filters)

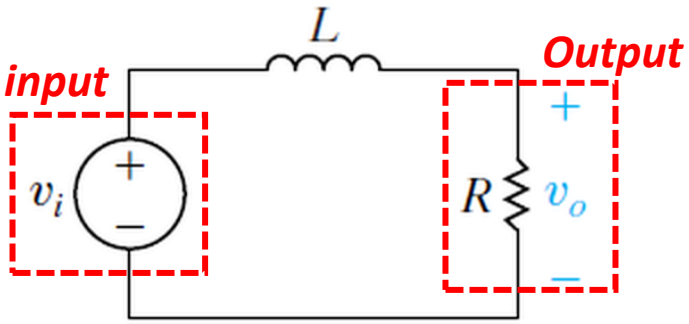
- **Magnitude plot:** |H(jω)| versus ω
- **Phase angle plot:** θ(jω) versus ω

- **Passband:** frequencies passed from the input to the output
- **Stopband:** frequencies not in the passband
- **Cutoff frequency (ω_c):** separates the passband and the stopband

Low-Pass Filter: Series RL Circuit

Series RL circuit

- **Input:** sinusoidal voltage source with varying frequency, v_i
- **Output:** voltage across the resistor, v_o



- **Low frequency (ωL ≪ R):** the inductor acts as a short circuit
- **High frequency (ωL ≫ R):** the inductor blocks the flow of current

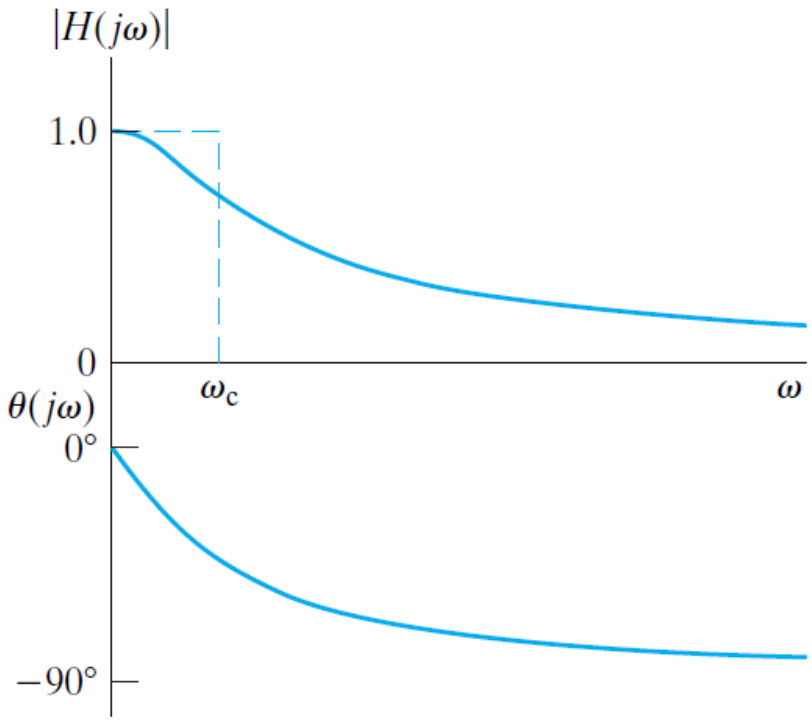
s-domain impedance Z: V(s) = Z I(s):
R → R,
L → sL
C → 1/sC

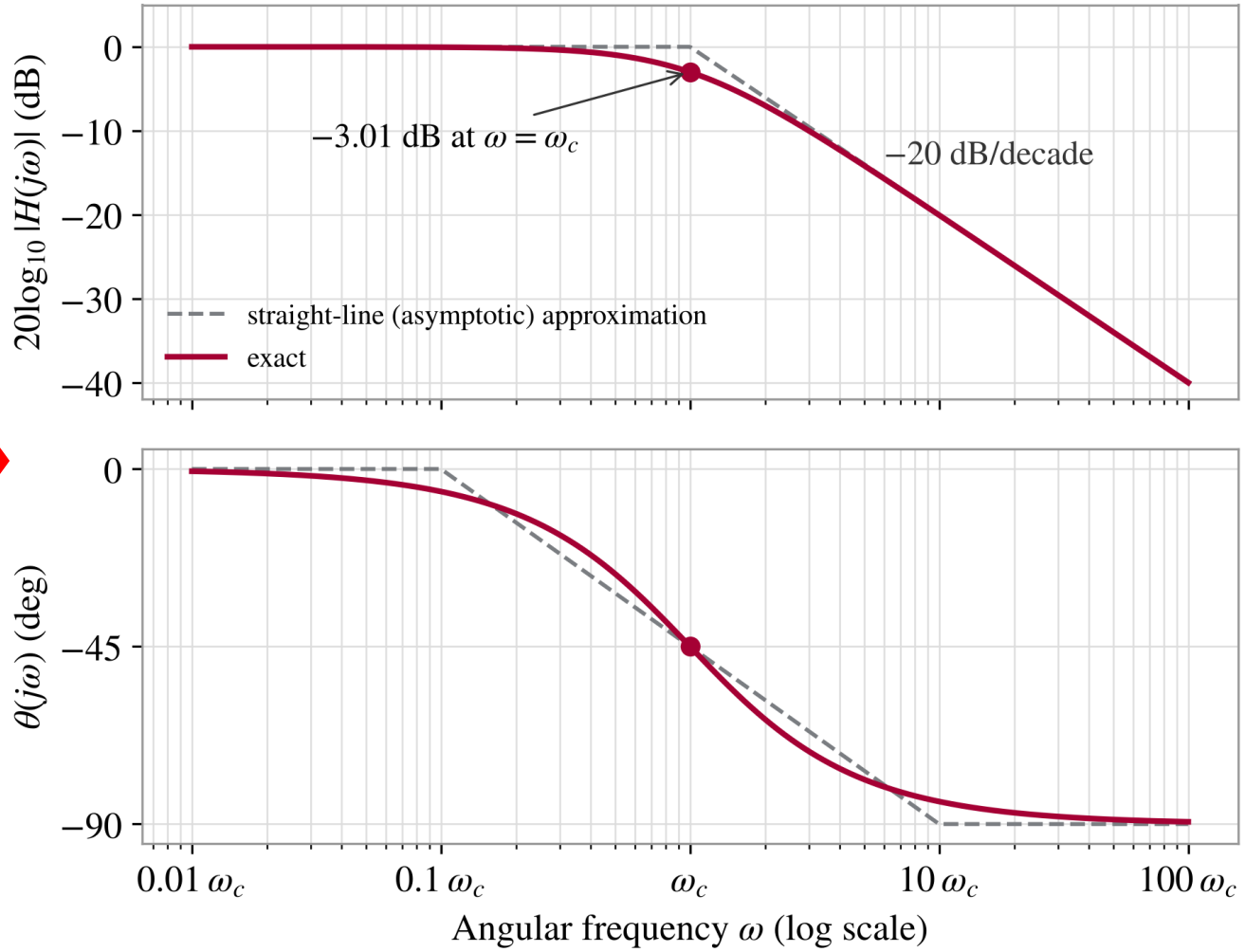
Transfer function between v_i and v_o

$$H(s) = \frac{V_o}{V_i} = \frac{R}{R + sL} = \frac{1}{s(L/R) + 1} = \frac{\omega_c}{s + \omega_c}, \qquad \omega_c = \frac{R}{L}$$

$$|H(j\omega)| = \frac{R/L}{\sqrt{\omega^2 + (R/L)^2}}, \qquad \theta(j\omega) = -\tan^{-1}\left(\frac{\omega L}{R}\right)$$

Low-Pass Filter: Frequency Response

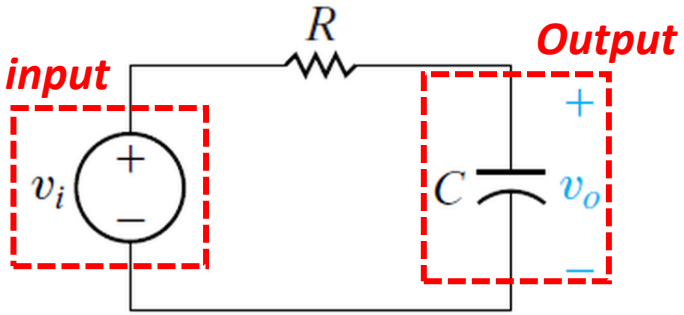




Low-Pass Filter: Series RC Circuit

Series RC circuit

- **Input:** sinusoidal voltage source with varying frequency, v_i
- **Output:** voltage across the capacitor, v_o



- **Low frequency ($1/\omega C \gg R$):** the capacitor blocks current → high voltage across C
- **High frequency ($1/\omega C \ll R$):** the capacitor acts as a short circuit → low voltage across C

Transfer function between v_i and v_o

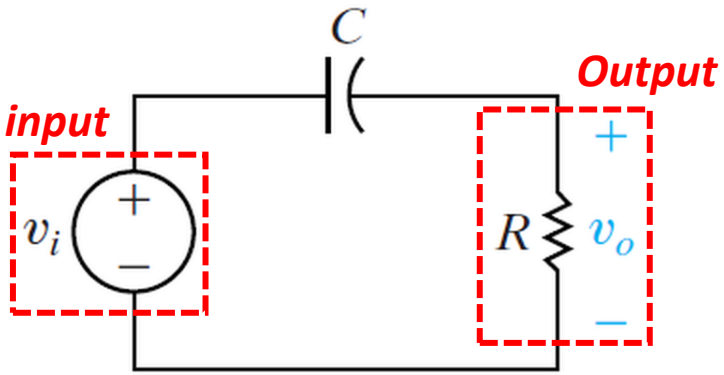
$$H(s) = \frac{V_o}{V_i} = \frac{1/sC}{R + 1/sC} = \frac{1}{s(RC) + 1} = \frac{\omega_c}{s + \omega_c}, \qquad \omega_c = \frac{1}{RC}$$

$$|H(j\omega)| = \frac{1/RC}{\sqrt{\omega^2 + (1/RC)^2}}, \qquad \theta(j\omega) = ?$$

High-Pass Filter: Series RC Circuit

Series RC circuit

- Input: sinusoidal voltage source with varying frequency, v_i
- Output: voltage across the resistor, v_o

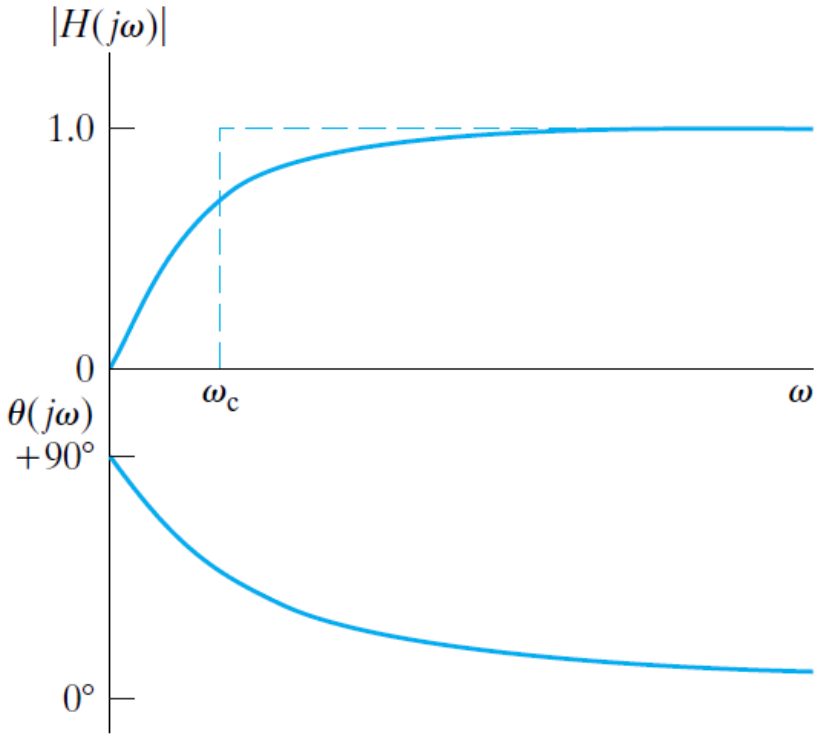


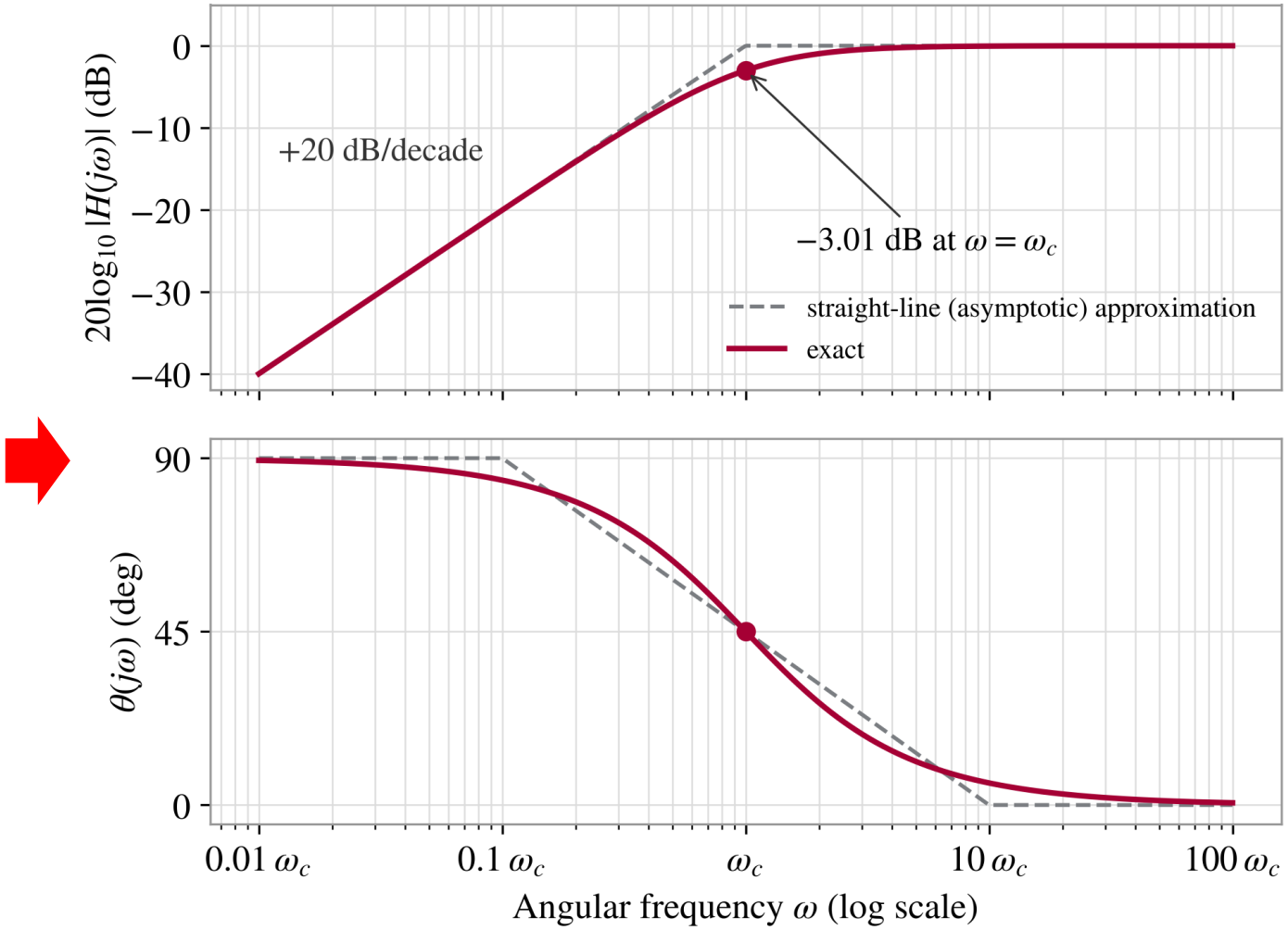
Transfer function between v_i and v_o

$$H(s) = \frac{V_o}{V_i} = \frac{R}{R + 1/sC} = \frac{s}{s + 1/RC} = \frac{s}{s + \omega_c}$$

$$|H(j\omega)| = \frac{\omega}{\sqrt{\omega^2 + (1/RC)^2}}, \qquad \theta(j\omega) = \tan^{-1}\left(\frac{1}{\omega RC}\right)$$

High-Pass Filter: Frequency Response

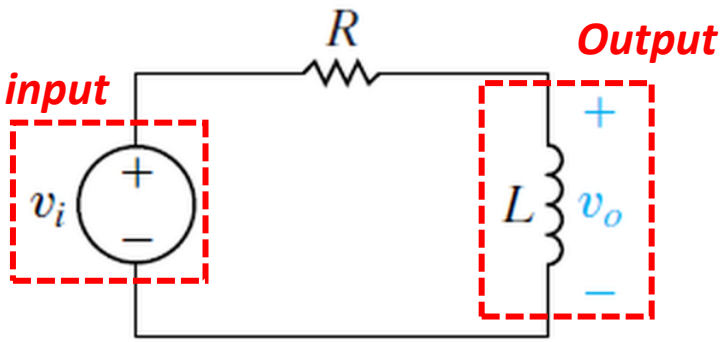




High-Pass Filter: Series RL Circuit

Series RL circuit

- **Input:** sinusoidal voltage source with varying frequency, v_i
- **Output:** voltage across the inductor, v_o



- Transfer function between v_i and v_o

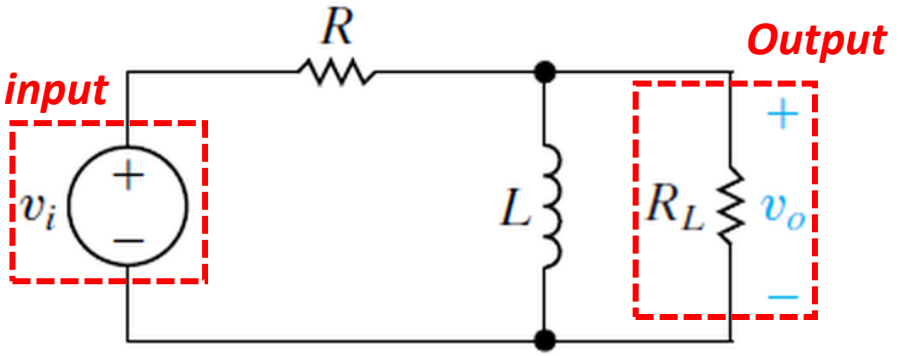
$$H(s) = \frac{V_o}{V_i} = \frac{sL}{R + sL} = \frac{s}{s + R/L} = \frac{s}{s + \omega_c}$$

$$|H(j\omega)| = \frac{\omega}{\sqrt{\omega^2 + (R/L)^2}}, \qquad \theta(j\omega) = ?$$

Passive Filter Design with a Load

Series RL (high-pass) + load

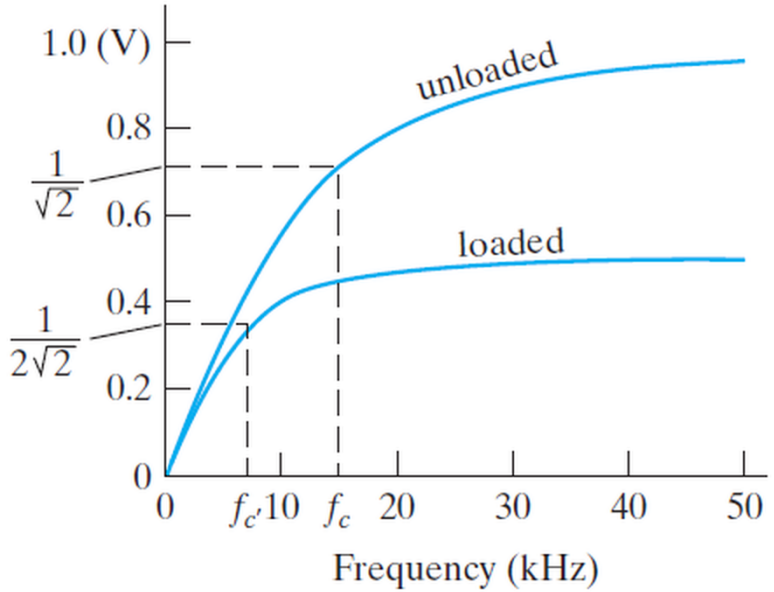
- **Input:** sinusoidal voltage source with varying frequency, v_i
- **Output:** voltage across the load, v_o



Transfer function between v_i and v_o

$$H(s) = \frac{V_o}{V_i} = \frac{\dfrac{R_L sL}{R_L + sL}}{R + \dfrac{R_L sL}{R_L + sL}} = \frac{\left(\dfrac{R_L}{R + R_L}\right)s}{s + \left(\dfrac{R_L}{R + R_L}\right)\dfrac{R}{L}} = \frac{Ks}{s + \omega_c}$$

$$K = \frac{R_L}{R + R_L}, \qquad \omega_c = K\frac{R}{L}$$



R = R_L = 500 Ω, L = 5.31 mH

Dilemma: The **load decreases** both the **gain** and the **cutoff frequency**

Active filters keep the filter's T.F. unchanged

Disadvantages of Passive Filters

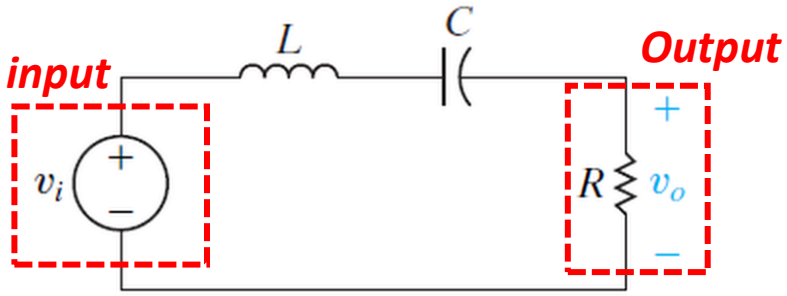
- The largest output amplitude is usually 1 (except the RLC resonant filter)
- Placing an impedance in series with the source or in parallel with the load decreases this amplitude

Active filters (op-amp based) isolate the load and can provide gain, so the transfer function does not change with the load

Bandpass Filter: Series RLC Circuit

Series RLC circuit

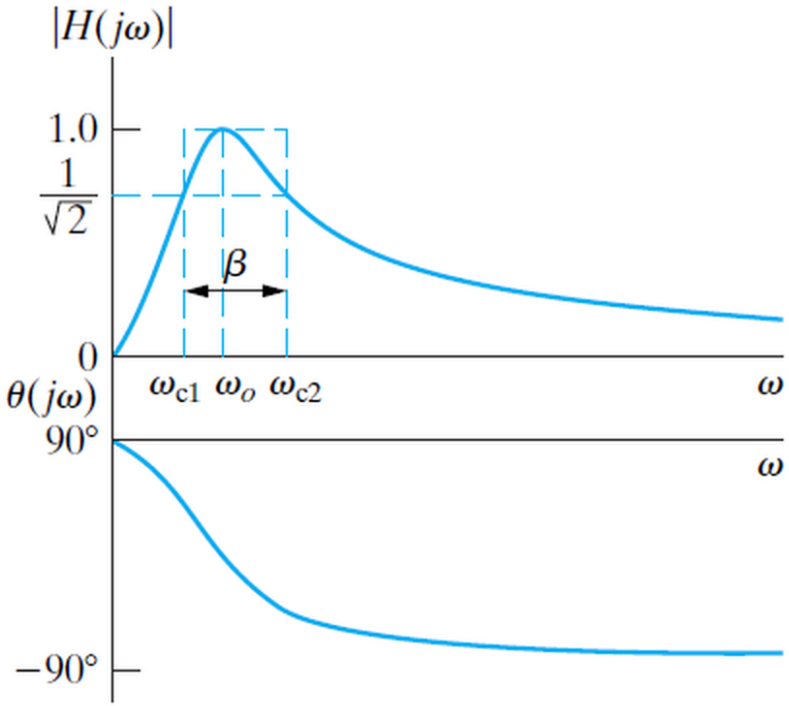
- **Input:** sinusoidal voltage source with varying frequency, v_i
- **Output:** voltage across the resistor, v_o



- **Low frequency (ω = 0):** capacitor open, inductor short → no current → output zero
- **High frequency (ω → ∞):** capacitor short, inductor open → no current → output zero

- How about ω between 0 and ∞?

$$H(s) = \frac{(R/L)\,s}{s^2 + (R/L)\,s + 1/LC}$$



Bandpass Filter: Center Frequency, Bandwidth, Q

$$\omega_o = \sqrt{\frac{1}{LC}}$$

$$\omega_{c1} = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

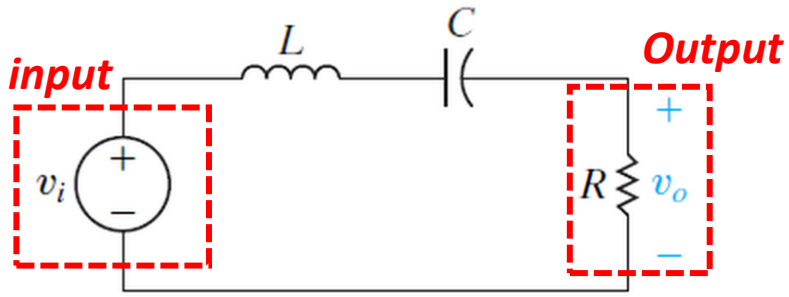
$$\omega_{c2} = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

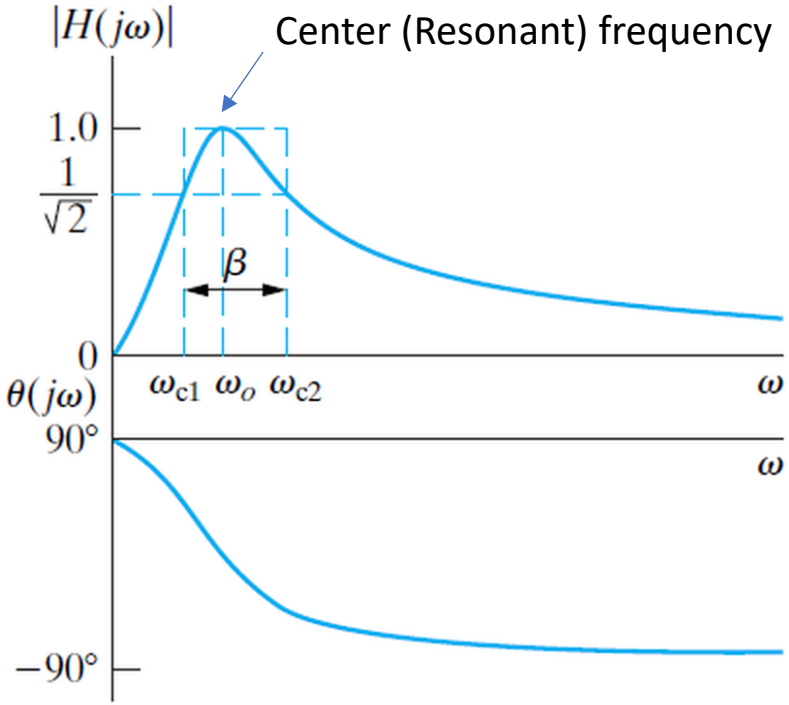
$$\omega_o = \sqrt{\omega_{c1}\,\omega_{c2}} = \sqrt{1/LC}$$

$$H_{\max} = |H(j\omega_o)| = 1$$

$$\beta = \omega_{c2} - \omega_{c1} = R/L$$

$$Q = \omega_o/\beta = \sqrt{L/(CR^2)}$$

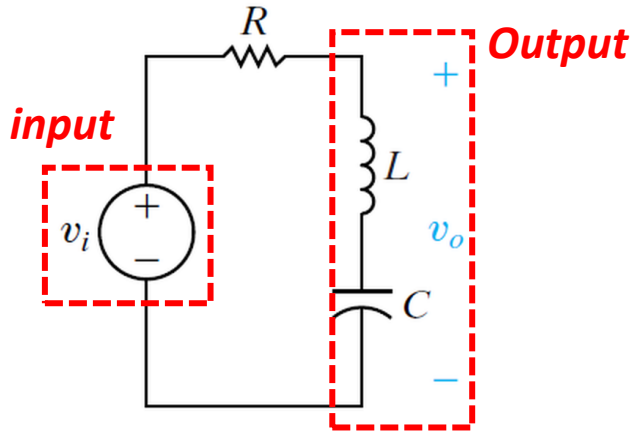




Bandreject Filter: Series RLC Circuit

Series RLC circuit

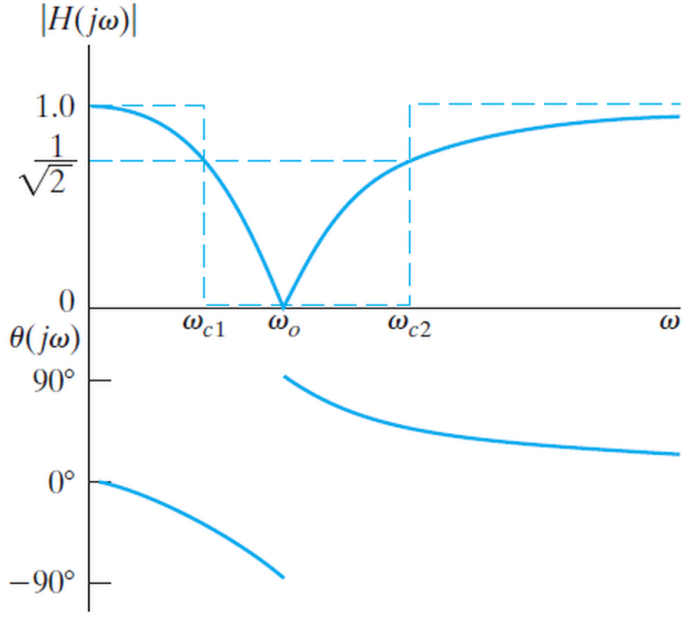
- **Input:** sinusoidal voltage source with varying frequency, v_i
- **Output:** voltage across the inductor and capacitor, v_o



$$H(s) = \frac{s^2 + 1/LC}{s^2 + (R/L)\,s + 1/LC}$$

- Bandpass and bandreject filters are complementary:

$$H_{\mathrm{BP}}(s) + H_{\mathrm{BR}}(s) = \frac{(R/L)s + s^2 + 1/LC}{s^2 + (R/L)s + 1/LC} = 1$$

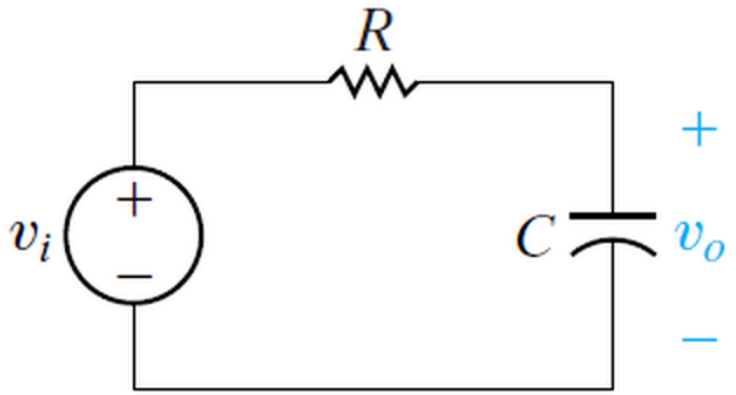


Design: 1st-Order RC Low-Pass Filter

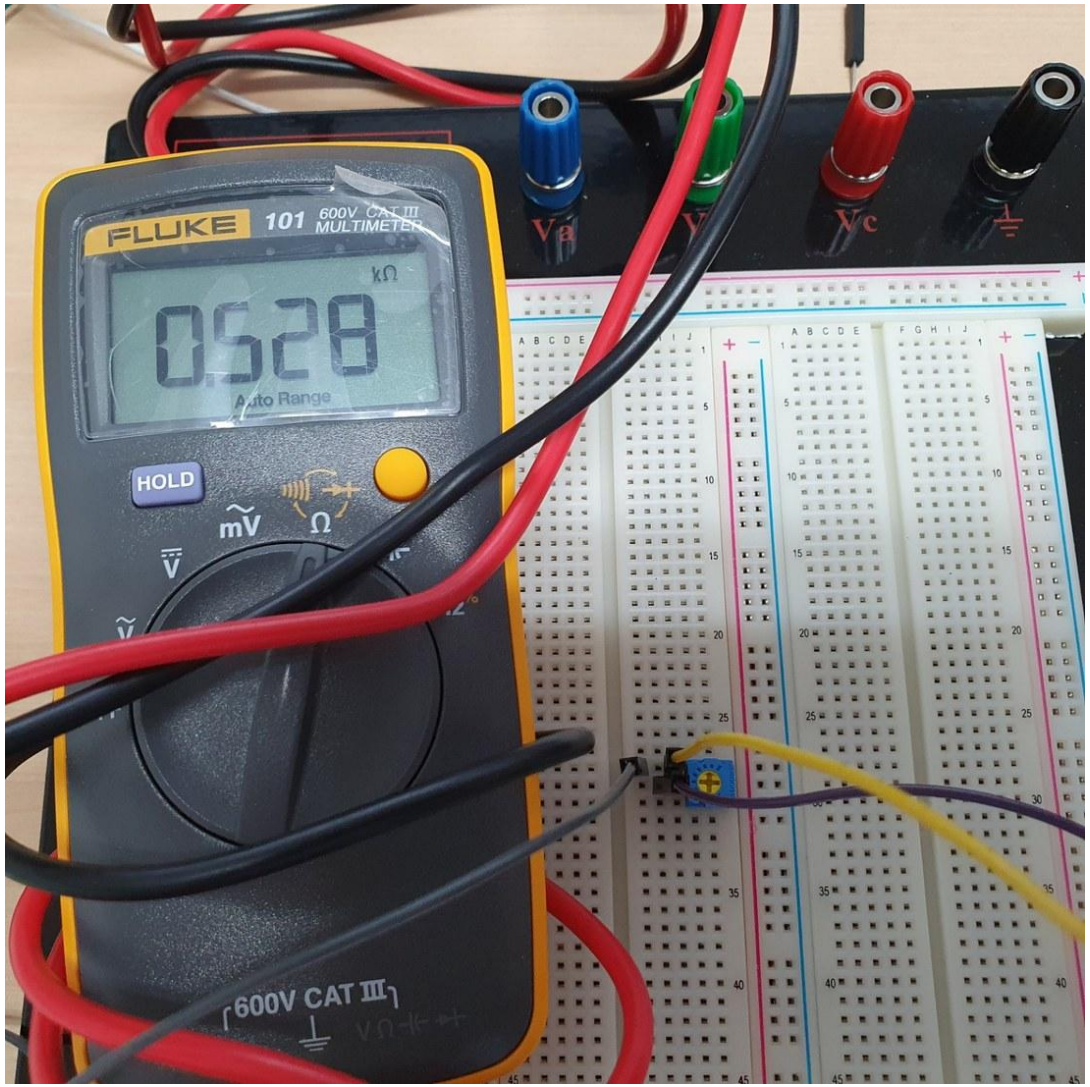
- **Goal:** build an RC low-pass filter and measure its frequency response with NI myDAQ

Cutoff: $f_c \approx 30$ Hz

- Choose C, then set R with a trimmer potentiometer (check it with the multimeter)

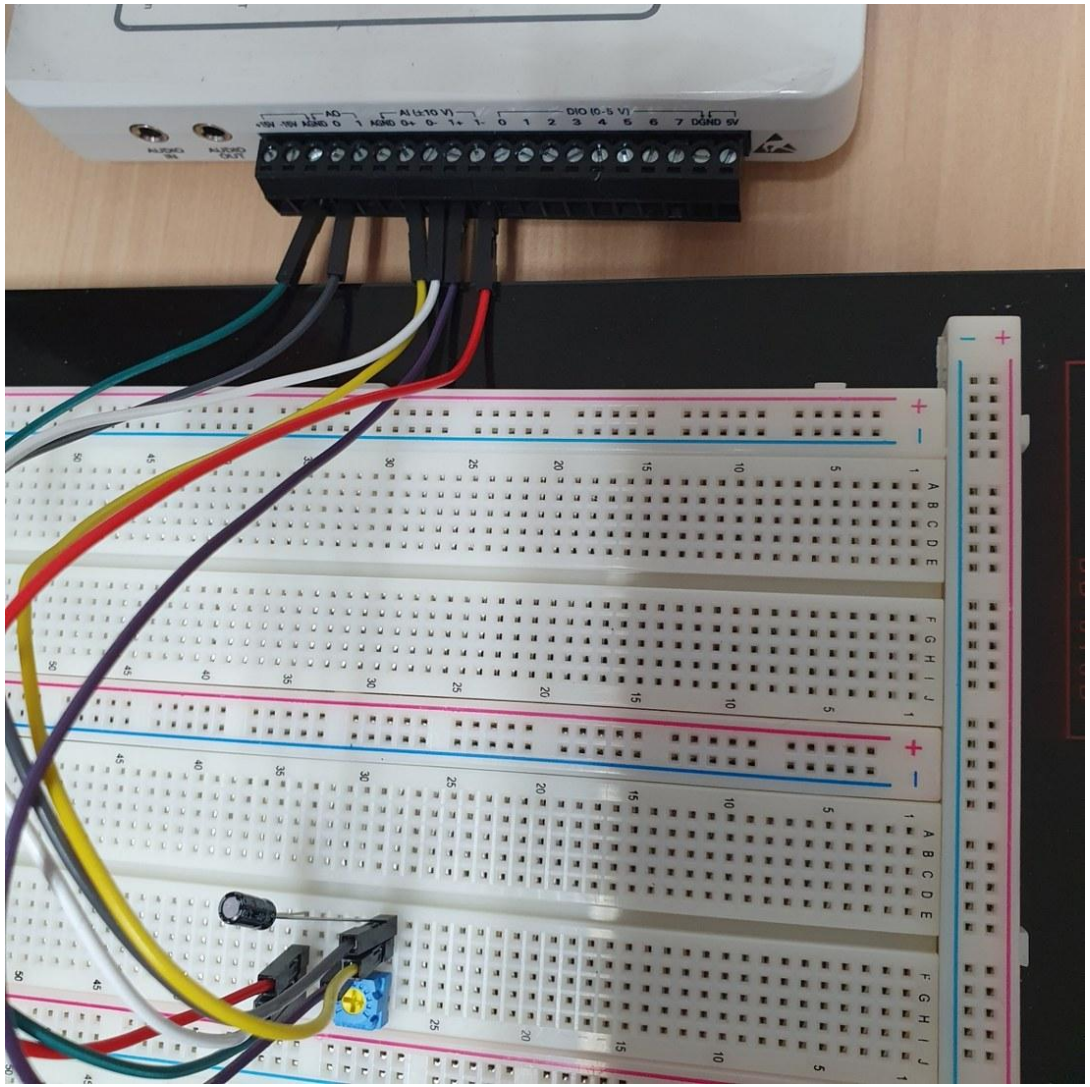


Setting R with the Multimeter



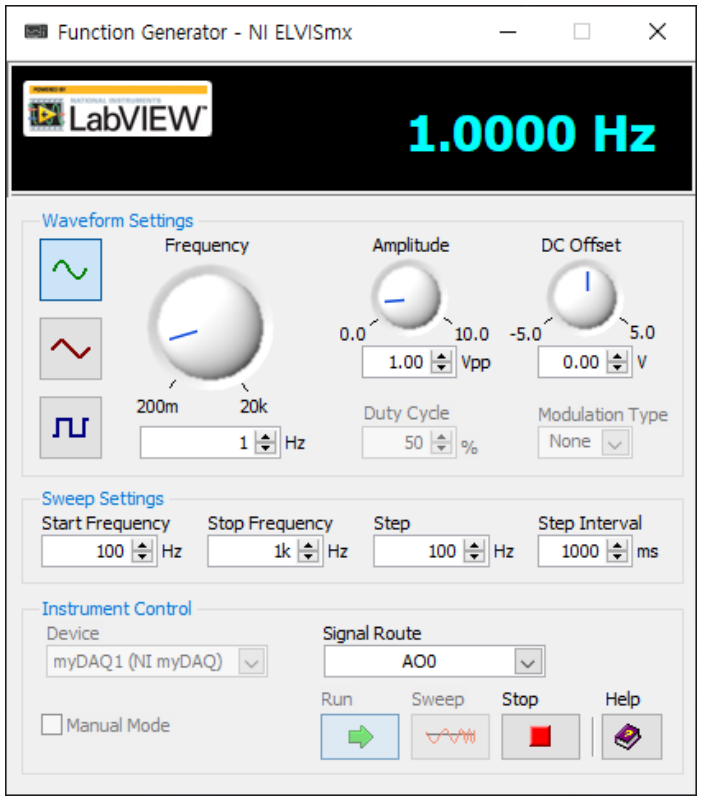
- Measure the trimmer potentiometer resistance with the digital multimeter before wiring the circuit
  - Fluke 101, auto range, kΩ

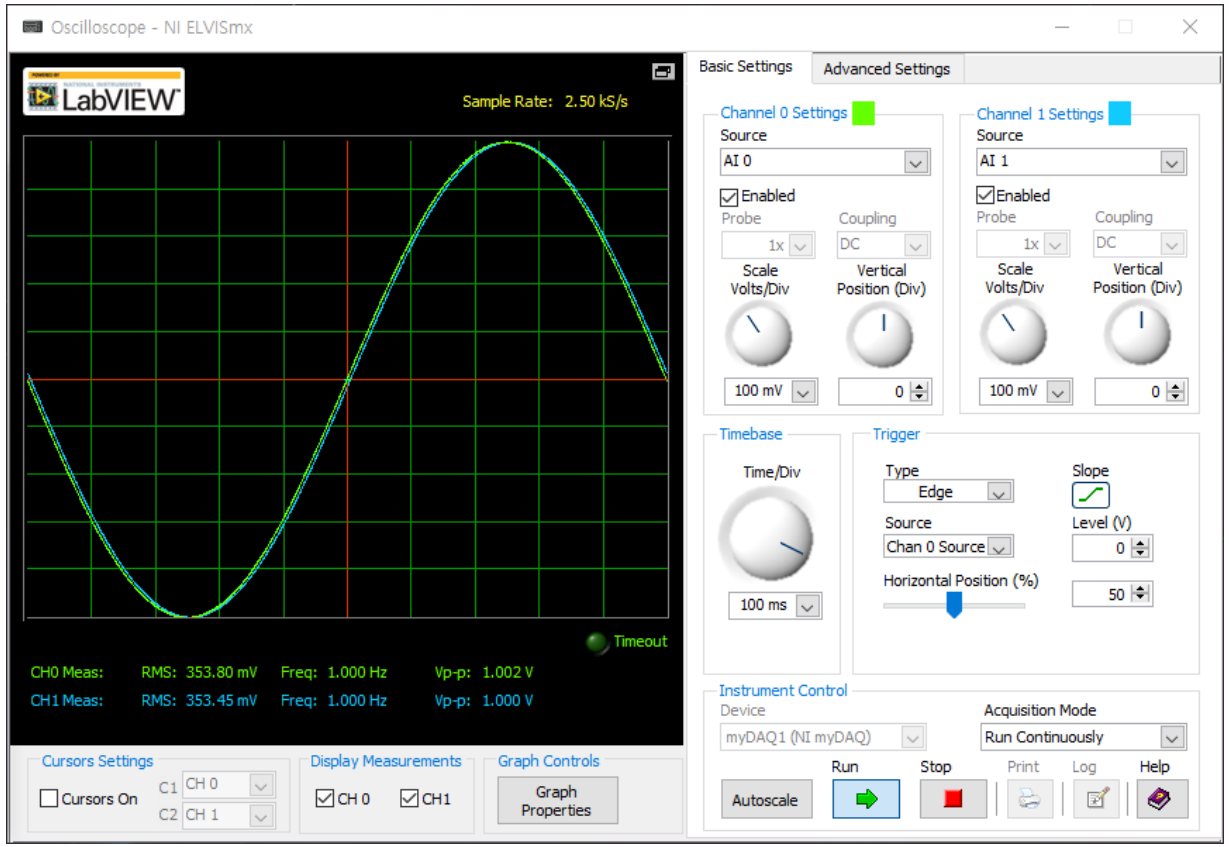
Wiring the Circuit with NI myDAQ



- Build the RC circuit on the breadboard connected to the NI myDAQ terminal block
- Function generator output (AO0) drives the circuit input
- Oscilloscope: CH0 = AI0 (input v_i), CH1 = AI1 (output v_o across C)

Measurement at f = 1 Hz

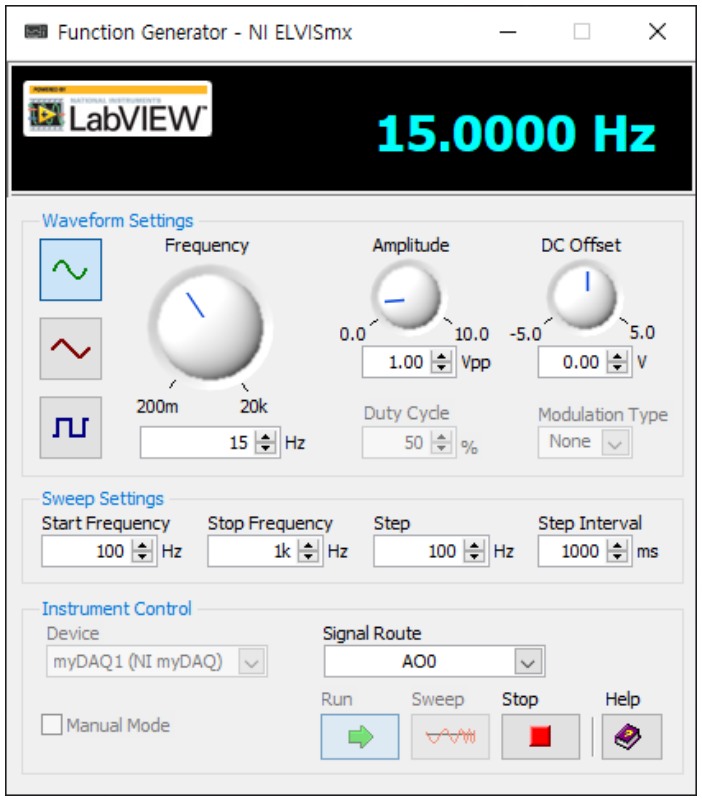


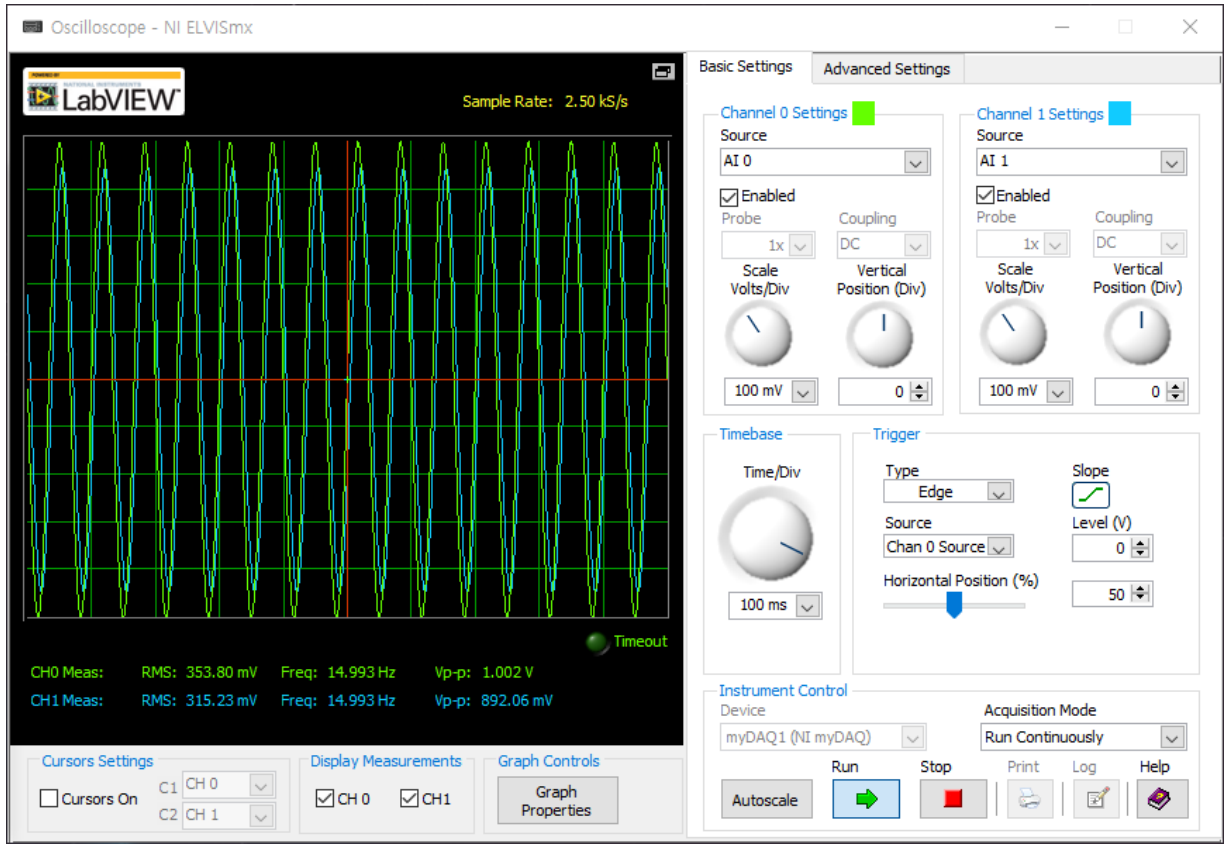


CH0 (green) = input v_i, CH1 (blue) = output v_o

f = 1 Hz: V_o/V_i = 353.45/353.80 mV (RMS) = 1.00, passband

Measurement at f = 15 Hz

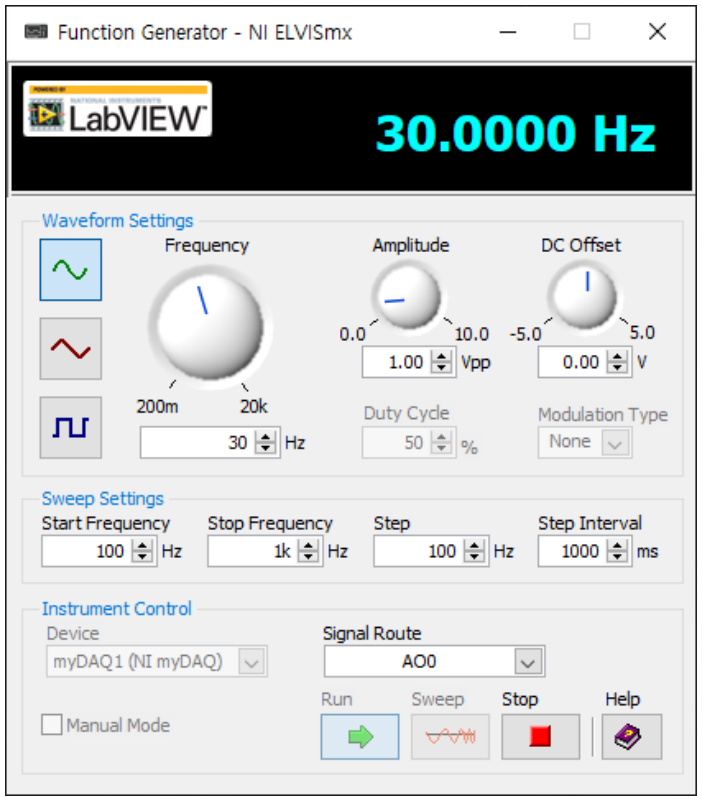


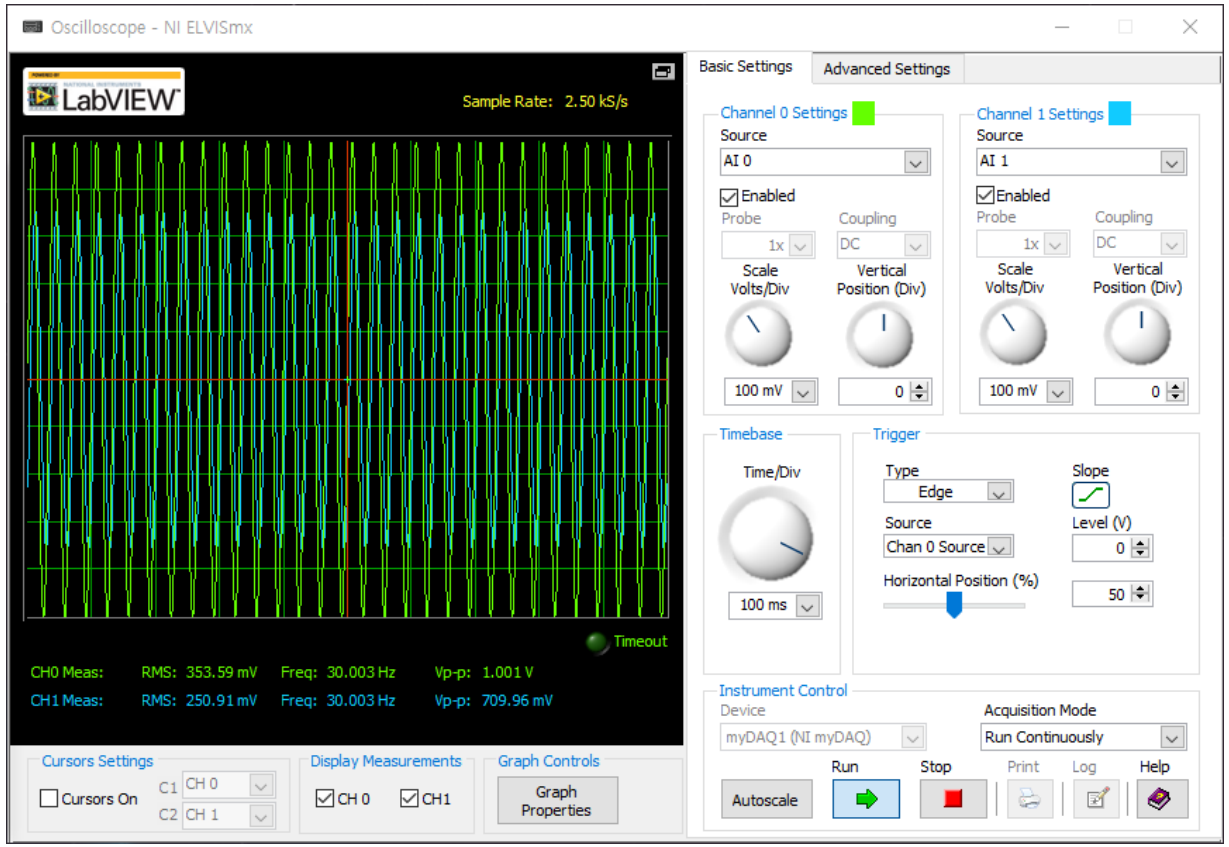


CH0 (green) = input v_i, CH1 (blue) = output v_o

f = 15 Hz: $V_o/V_i = 315.23/353.80 = 0.89$ (−1.0 dB)

Measurement at f = 30 Hz

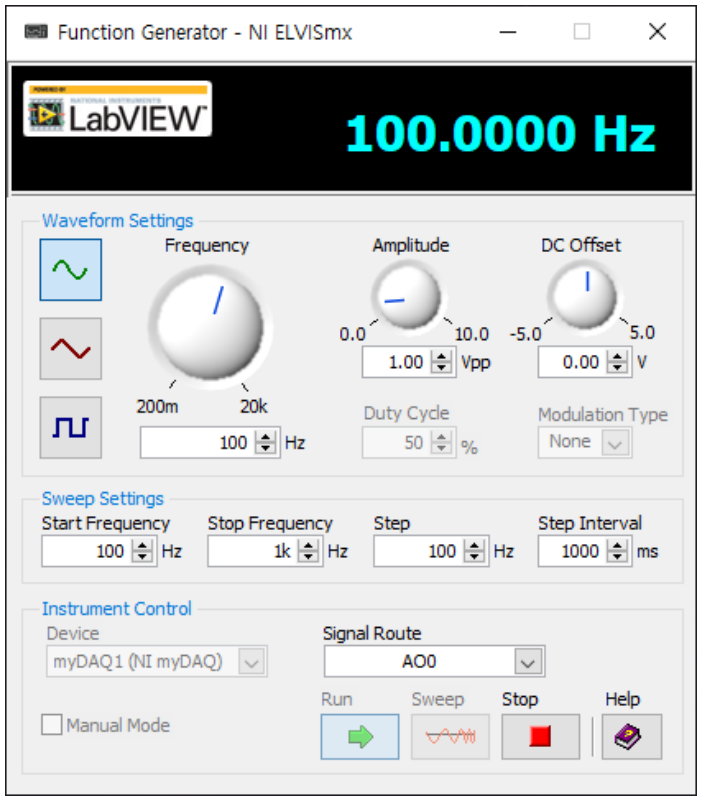


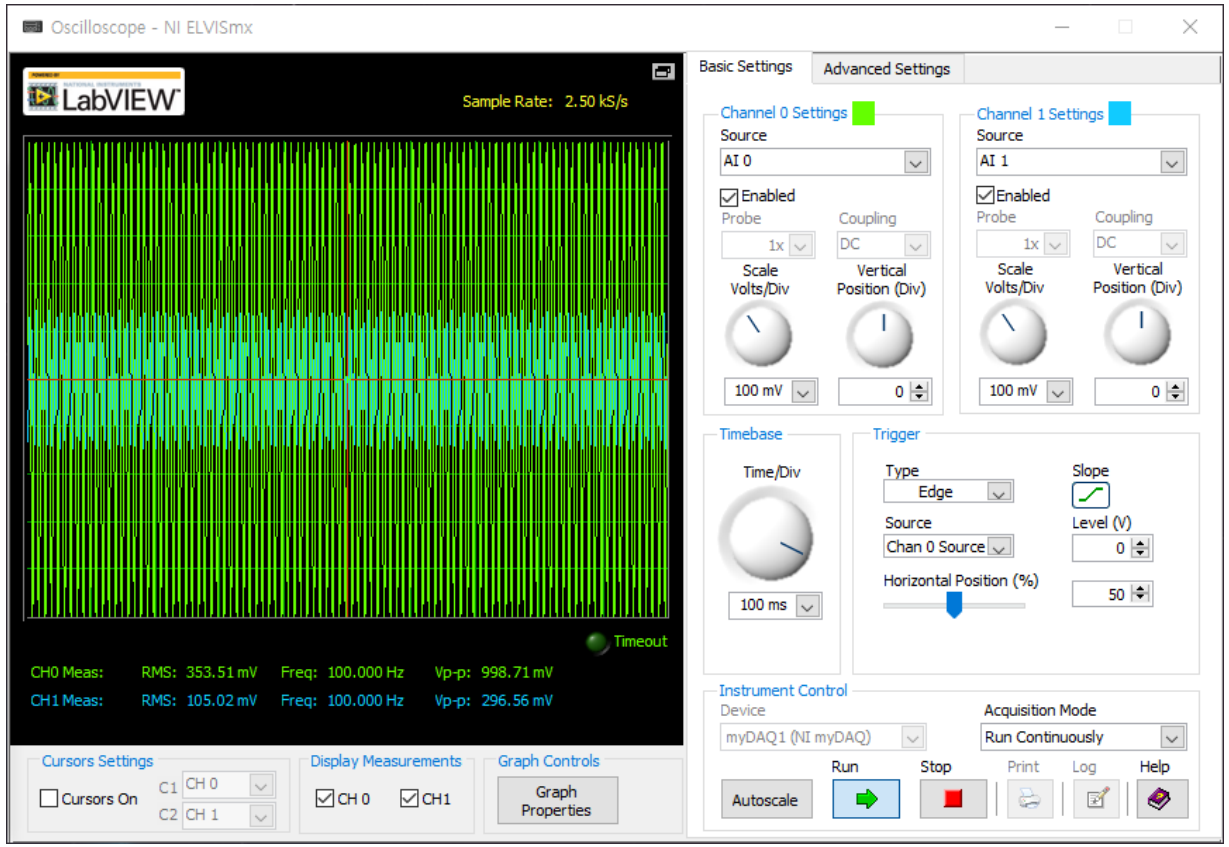


CH0 (green) = input v_i, CH1 (blue) = output v_o

f = 30 Hz: $V_o/V_i = 250.91/353.59 = 0.71$ (−3.0 dB), close to the cutoff

Measurement at f = 100 Hz

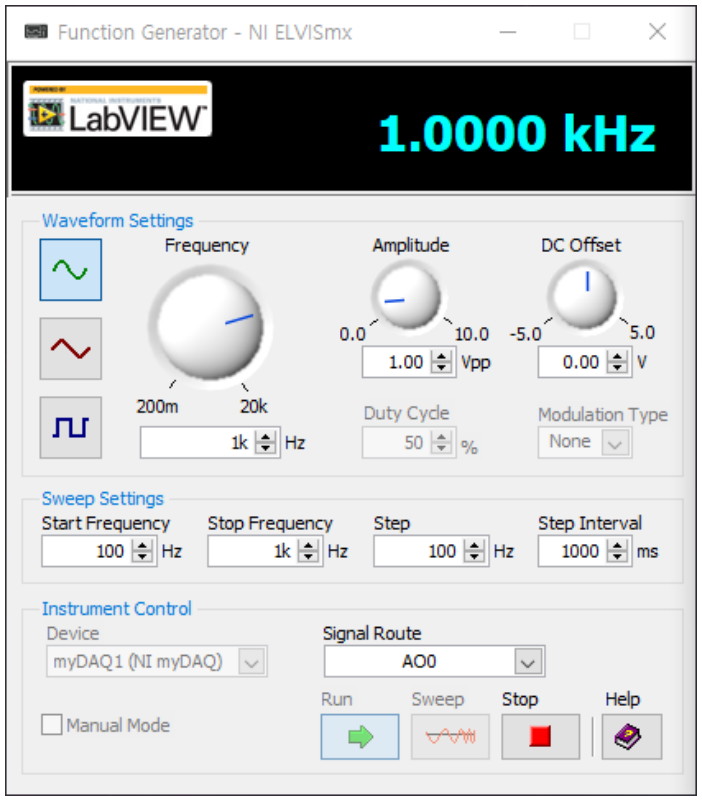


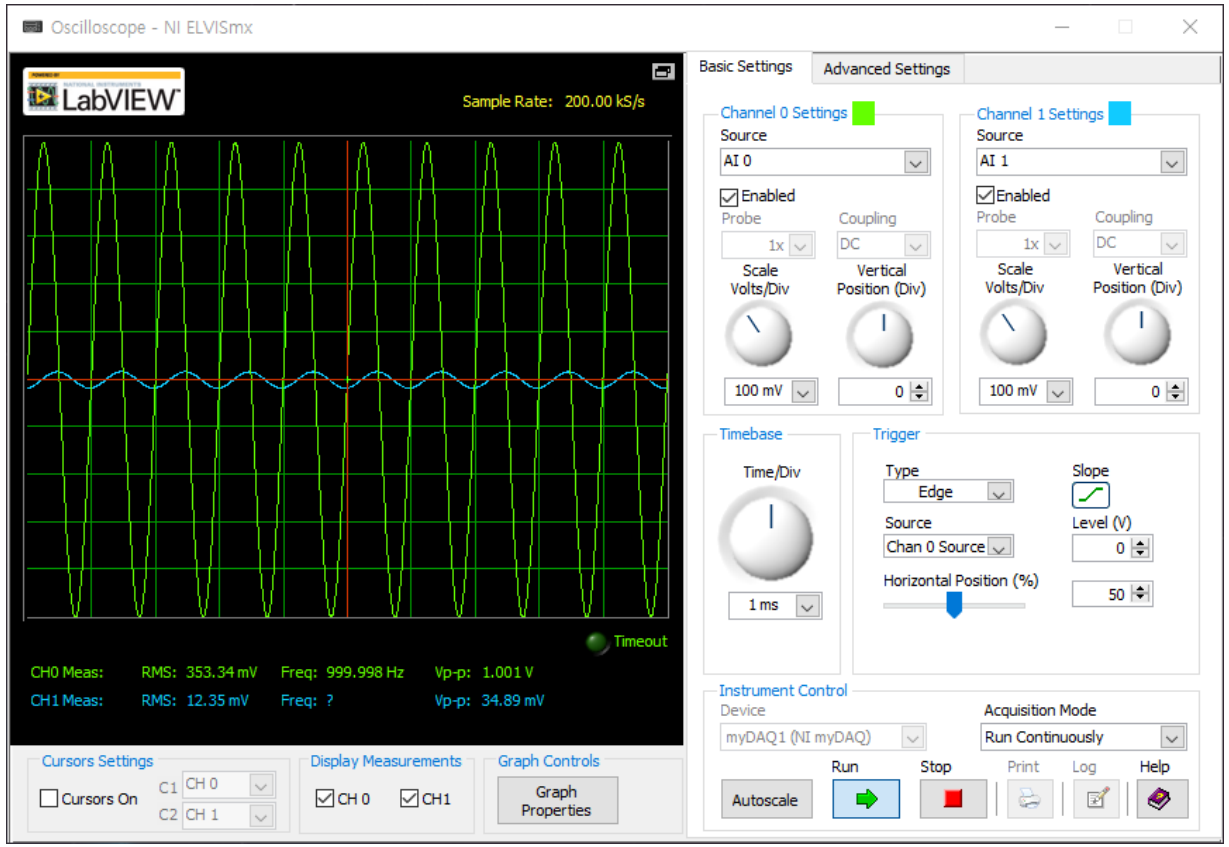


CH0 (green) = input v_i, CH1 (blue) = output v_o

f = 100 Hz: $V_o/V_i = 105.02/353.51 = 0.30$ (−10.5 dB)

Measurement at f = 1 kHz

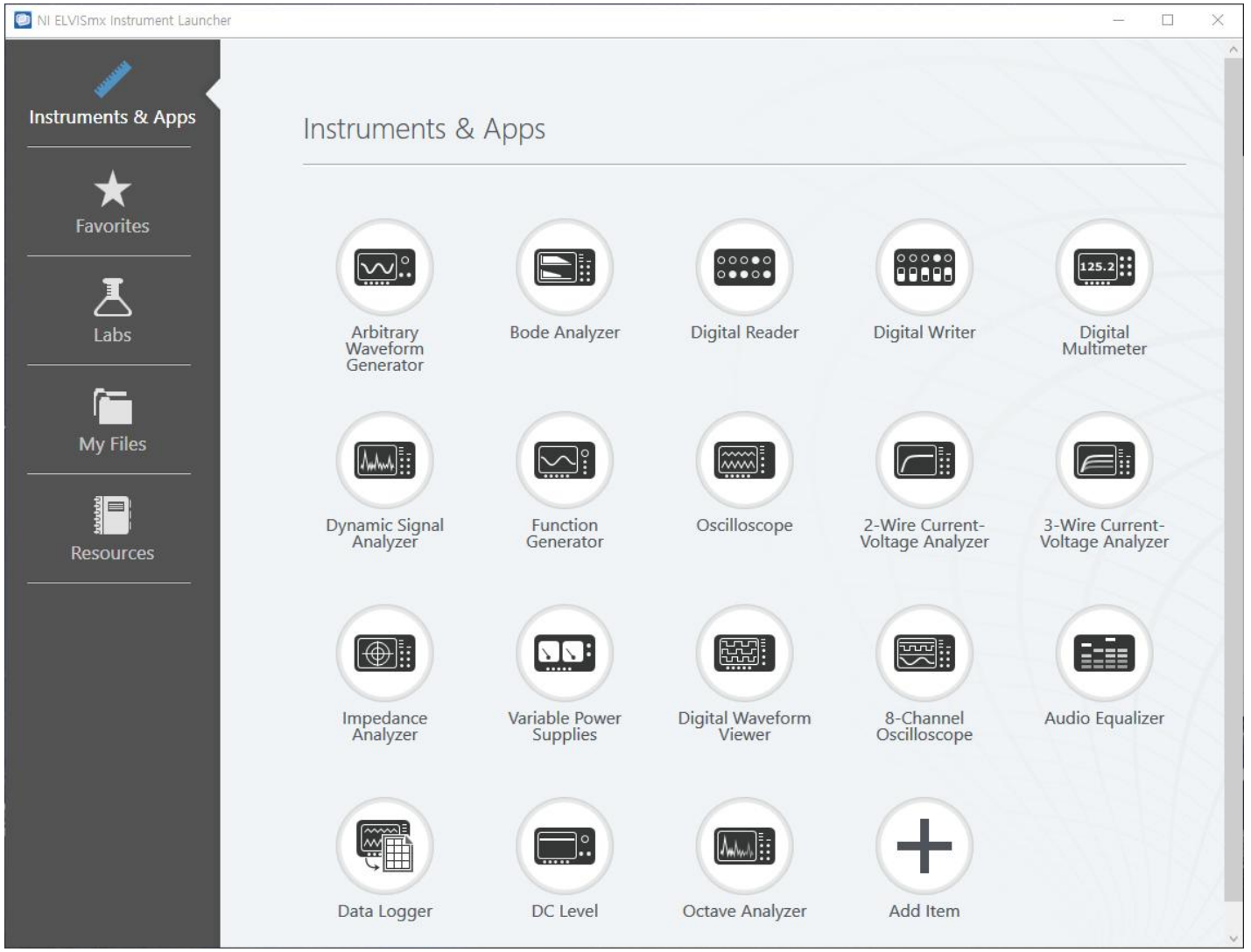




CH0 (green) = input v_i, CH1 (blue) = output v_o

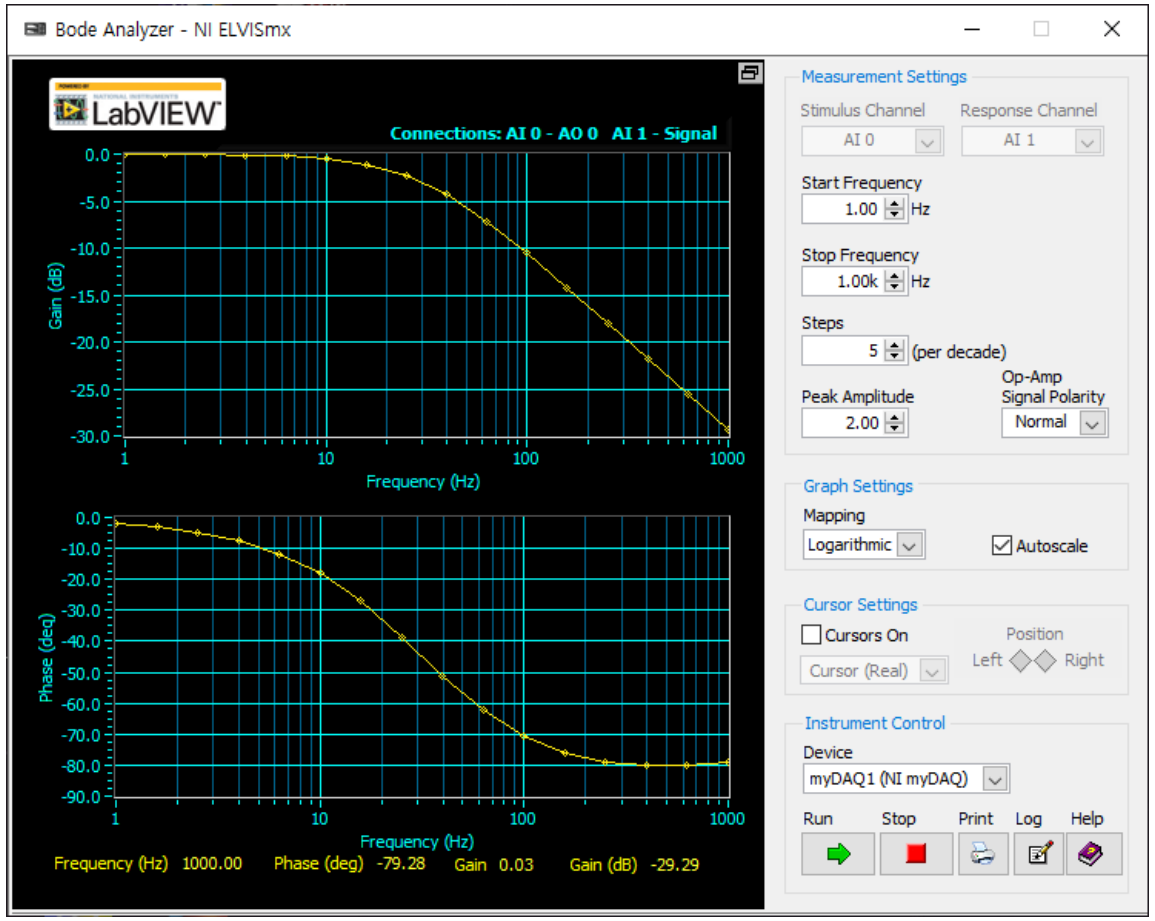
f = 1 kHz: $V_o/V_i = 12.35/353.34 = 0.035$ (−29.1 dB)

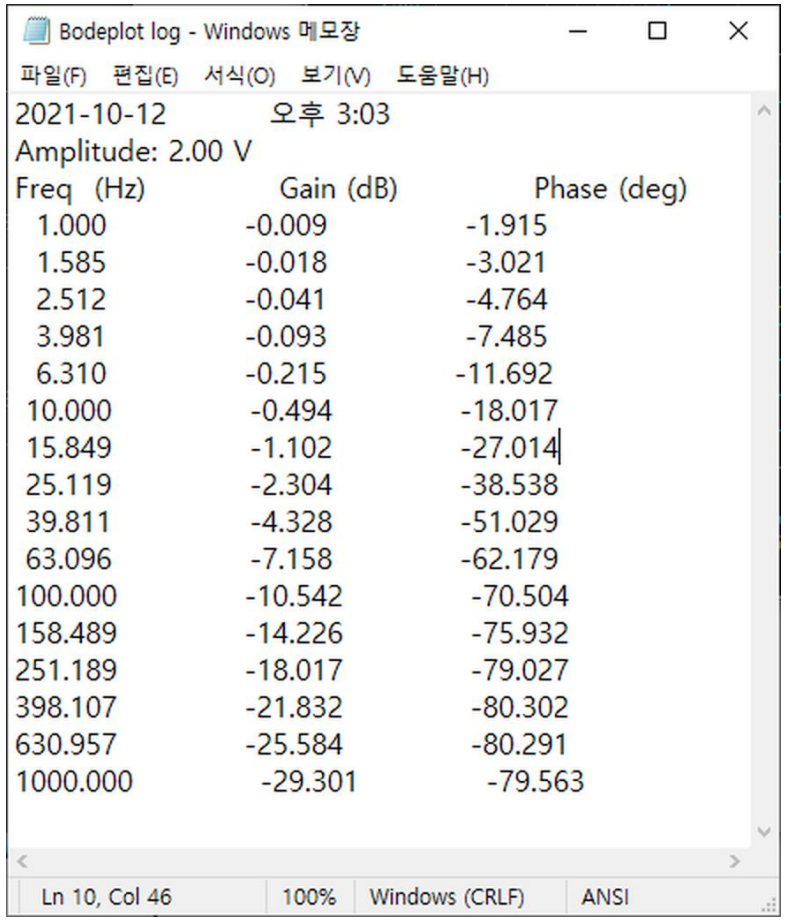
NI ELVISmx Instrument Launcher



- Open the Bode Analyzer from the Instrument Launcher
- It sweeps the stimulus frequency and measures gain and phase automatically

Bode Analyzer Result





−3 dB between 25 Hz (−2.30 dB) and 40 Hz (−4.33 dB); at 1 kHz: −29.3 dB, −79.6°

Report 2

1. Circuit design with a 1st-order passive low-pass filter

- A. Build an RC circuit with a cutoff frequency of 1 kHz. Measure $|H(j\omega_c)|$ at the cutoff frequency and compare it with the theoretical value
- B. Use the Bode Analyzer of the NI ELVISmx Instrument Launcher to obtain the Bode plot of the circuit in A (start frequency: 1 Hz, stop frequency: 20 kHz)
- C. Plot the theoretical Bode plot of the transfer function of the circuit in A together with the result of B for comparison, using MATLAB (+5 pt) or Excel
  - Gain: y-axis in dB, x-axis in Hz on a log (decade) scale. Phase: y-axis in degrees, x-axis in Hz on a log (decade) scale

2. Repeat the above procedure using an RL circuit